

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		•CH ₂ COOH			1	1		
		(ii)	I	CH ₃ COOH → ClCH ₂ COOH 60.0 → 94.5 89.0 → 140.2 (1) increase in mass 140.2 – 89.0 = 51.2 g (1) accept 51 g		1	1	2		
			II	CH ₃ COOH → CH ₂ (NH ₂)COOH 60.0 → 75.0 89.0 → 111.3 (1) percentage yield = $\frac{49.2 \times 100}{111.3} = 44.3$ (1) answer must be to 3 significant figures		1 1		2	1	
			III	it exists as zwitterions / ionic compound (1) accept correctly drawn correct formula of zwitterion ionic compounds are not (generally) soluble in covalent solvents (1)	1		1	2		
	(b)			it does not contain a chiral centre / asymmetric carbon atom	1			1		

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	(c)			NH₂ H C H O N H H C H COOH accept zwitterion version		1		1		
	(d)			stoichiometric ratio is 1:1 (1) using pV = nRT volume = $\frac{0.300 \times 8.31 \times 373}{9.8 \times 10^4}$ (1) 9.49 dm³ (1) must be given in dm³	1					
	(e)			the (sodium) salt of the acid is formed		1		1		1
				Question 8 total	3	7	3	13	3	1